

Appearance outliers by mutual reachability of the team clusters

This section replaces the two-channel outlier test of the previous formulation (the centroid threshold of Eq. (2.15) and the DBSCAN noise channel) with a single rule. The rule keeps the robust median-based statistics of Eqs. (2.13)–(2.14): applied to the centroid distances, they define the certain core of each team; applied locally, per tracklet, they define an adaptive neighbourhood radius. An outlier is a tracklet that cannot be linked to either team cluster through a chain of mutually compatible tracklets.

Setup and notation

Each tracklet i of a sequence is represented by its embedding $z_i \in \mathbb{R}^d$, the coordinate-wise median of its crop vectors as defined in Eq. (2.13). Let n be the number of tracklets in the sequence. The dissimilarity between two tracklets is the Euclidean distance between their embeddings,

$$\delta_{ij} = \|z_i - z_j\|, \quad i \neq j. \quad (2.15)$$

For a fixed tracklet i , the distances to all other tracklets are sorted in increasing order and denoted

$$\delta_{i(1)} \leq \delta_{i(2)} \leq \dots \leq \delta_{i(n-1)}, \quad (2.16)$$

so that $\delta_{i(r)}$ is the distance from tracklet i to its r -th nearest neighbour. The tracklet embeddings are clustered by k -means into the two teams; c_0 and c_1 denote the centroids of team 0 and team 1 respectively, and T_0 and T_1 denote the sets of tracklets assigned to each team.

The team clusters

The reference populations of the rule are not the k -means assignments themselves. The assignment places every tracklet in one of the two teams, including the tracklets the rule is meant to flag, so membership in T_0 or T_1 cannot serve as evidence of legitimacy. Instead, the median rule of Eq. (2.14) is applied to the distances from each team's tracklets to their centroid, and the cluster of a team is the set of tracklets within the resulting radius. For team $b \in \{0, 1\}$, let $\delta_i^{(b)} = \|z_i - c_b\|$ for $i \in T_b$, and set

$$\mu^{(b)} = \text{median}_{i \in T_b} \delta_i^{(b)}, \quad s^{(b)} = \text{median}_{i \in T_b} |\delta_i^{(b)} - \mu^{(b)}|. \quad (2.17)$$

Definition 2.1 (Team cluster). *The cluster of team b is the set of tracklets whose distance to the team's centroid is within the median rule,*

$$\mathcal{K}_b = \{ i \in T_b : \delta_i^{(b)} \leq \mu^{(b)} + s^{(b)} \}. \quad (2.18)$$

The k -means assignment supplies only the distances from which the statistics are computed; membership in the cluster is decided by the median rule. A goalkeeper group or a corrupted tracklet forced into T_b by the assignment lies far from c_b and is excluded from $\mathcal{K}_b$ from the start. The members of $\mathcal{K}_0$ and $\mathcal{K}_1$ are the typical, certain team members; they are inliers by definition.

Per-tracklet radius

Each tracklet receives its own neighbourhood radius, derived from the k tracklets nearest to it. Fix a neighbour count k ; we use $k = 4$, matching the minimum neighbourhood size of the

DBSCAN instance that this rule replaces. The local scale of tracklet i is summarised by the median of its k nearest-neighbour distances and by their median absolute deviation,

$$\mu_i = \text{median}\{\delta_{i(1)}, \dots, \delta_{i(k)}\}, \quad s_i = \text{median}_{1 \leq r \leq k} |\delta_{i(r)} - \mu_i|, \quad (2.19)$$

computed only from tracklet i 's own neighbourhood: μ_i is the typical distance at which i finds its close neighbours, and s_i is the typical variation of those distances. The radius of tracklet i is

$$R_i = \mu_i + s_i. \quad (2.20)$$

A tracklet surrounded by many similar-looking tracklets receives a small radius; a tracklet in a sparse region receives a large one. The same median-plus-deviation rule as in Eq. (2.17) is used, but evaluated locally rather than over a team, so no global neighbourhood radius has to be estimated.

Mutual compatibility

Two tracklets are compatible when each lies within the radius of the other. Since the distance between them is a single number, this two-sided requirement reduces to a comparison with the smaller of the two radii.

Definition 2.2 (Mutual compatibility). *Tracklets i and j are mutually compatible when*

$$\delta_{ij} \leq \min(R_i, R_j). \quad (2.21)$$

The minimum makes the relation symmetric in substance as well as in form: a tracklet in a sparse region, whose own radius is wide, cannot connect to a tracklet in a dense region unless it also fits inside that tracklet's small radius. A connection must be agreeable from both sides, which prevents sparse tracklets from attaching themselves to dense groups unilaterally.

Linking and the outlier rule

A tracklet is linked to a team cluster when a chain of mutually compatible tracklets leads from the tracklet to a member of that cluster.

Definition 2.3 (Link to a team cluster). *Tracklet i is linked to the cluster $\mathcal{K}_b$ when there exists a finite sequence of tracklets $i = j_0, j_1, \dots, j_p$ such that each consecutive pair j_{t-1}, j_t is mutually compatible in the sense of Eq. (2.21) and the last tracklet satisfies $j_p \in \mathcal{K}_b$. The case $p = 0$ is the tracklet's own membership, $i \in \mathcal{K}_b$.*

Definition 2.4 (Appearance outlier). *Tracklet i is an appearance outlier when it is linked to neither team cluster,*

$$\text{outlier}(i) \iff i \text{ is not linked to } \mathcal{K}_0 \text{ and not linked to } \mathcal{K}_1. \quad (2.22)$$

Equivalently: starting from tracklet i and stepping only between mutually compatible tracklets, an inlier eventually reaches a certain team member, while an outlier does not. The rule has a single free parameter, the neighbour count k ; the clusters, the radii, and the links are determined by the data.

Why the rule is correct for this task

The rule flags exactly the tracklets that the downstream positional rules are meant to receive. Three cases cover the possibilities.

An isolated tracklet. A lone referee in a kit unlike either team, or a corrupted tracklet whose median crop vector lies between the teams, is far from both centroids and therefore outside both clusters; and it is compatible with nothing, since every other tracklet either lies outside its radius or refuses the connection through its own smaller radius. No chain can start, so the tracklet is linked to neither cluster and is flagged.

A small sparse group. A handful of unusual tracklets may be compatible with one another, yet no member belongs to a cluster and no member is compatible with anything outside the group. Every chain that starts inside the group stays inside it, never reaching a cluster member, and the whole group is flagged.

A dense off-cluster group. This is the decisive case. Four or more goalkeeper tracklets, or several officials in matching kit, form a tight clump: each member has close neighbours, so any test based on local density alone accepts all of them. Under the present rule the clump lies far from both centroids, so none of its members belongs to $\mathcal{K}_0$ or $\mathcal{K}_1$; and chains started inside the clump run only between its members — their small radii reach one another but not across the gap to the teams — so no member is linked to either cluster, and the entire clump is flagged. The previous formulation needed a separate centroid threshold precisely for this case; here it is covered by the same linking criterion as the other two.

Distance to the teams is expressed by the links themselves. No separate “far from the team” threshold appears in the outlier rule, because the absence of a link already encodes it. A tracklet fails to reach a cluster exactly when, at some point between it and the certain team members, the gap exceeds what the radii can bridge. Being far from both teams and being unlinked from both clusters are the same statement. The centroid test of the previous formulation is not discarded: it survives as the cluster definition of Eq. (2.18), which determines where every successful chain must end.

Limitation: chaining

The linking relation is transitive, and this is the cost of the rule. A tracklet that is not directly compatible with any certain team member can still be linked through a chain of intermediaries, each consecutive pair satisfying Eq. (2.21). A corrupted tracklet lying near a noisy but legitimate player tracklet, which in turn lies near the team, may ride such a chain to a cluster member and escape flagging. This is the chaining effect familiar from single-linkage clustering. Two properties bound its impact in this pipeline. First, every step of a chain must be mutual: each link has to fit inside the smaller of the two radii involved, so a chain cannot be built out of one-sided reaches through sparse space. Second, the flagged set is not the final decision; the positional rules applied afterwards re-examine the outliers, so the consequence of a tracklet that escapes flagging is confined to that later stage.

Degenerate cases

Two safeguards keep the statistics meaningful. When the sequence contains too few tracklets, $n \leq k+1$, the local statistics of Eq. (2.19) are computed from fewer than k genuine neighbours and the rule is skipped; no tracklet is flagged by this test. When the k nearest-neighbour distances of a tracklet are nearly identical, the local deviation s_i can collapse towards zero and produce an unrealistically tight radius; s_i is therefore floored at a small positive constant so that a vanishing spread cannot, by itself, unlink a tracklet. The same floor is applied to $s^{(0)}$ and $s^{(1)}$ in Eq. (2.17), so that a team with nearly identical centroid distances does not shrink its cluster to the median alone.

Remark 2.1 (Relation to DBSCAN). *The rule replaces the DBSCAN noise channel of the previous formulation, and the relation between the two is exact at the level of the density test: a point has at least k neighbours within a radius ε if and only if its k -th nearest-neighbour distance is at most ε . The differences lie in how the radius is obtained and in what counts as legitimate. DBSCAN uses a single global radius, estimated from the data and scaled by a fixed factor, so the same cutoff is applied to dense and sparse regions alike; here each tracklet carries its own radius R_i , derived from its own neighbourhood by the same median rule as Eq. (2.14), and no radius parameter remains to be tuned. DBSCAN rescues border points one-sidedly: a point that fails the density test joins a cluster whenever it lies within ε of a core point. The present rule admits border tracklets only through the mutual condition of Eq. (2.21), so the rescue must be agreed to by the denser side. Finally, DBSCAN accepts any sufficiently dense region as a cluster, which is why a compact group of goalkeepers is invisible to it; the present rule requires a link to a team cluster defined by the median rule around each centroid, which is the domain knowledge the outlier step exists to enforce. The parameters reduce from two (ε and the minimum neighbourhood size) to one (the neighbour count k).*